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Proxmox Networking for VMware vSphere admins

December 28, 2023 by Brandon Lee



One of the challenges that we run into when we are more familiar with one vendor over another is the difference in the technologies, how they work for customers, what they are called, and how to configure them. On the networking side of things, this can be the case as well. If you are familiar with VMware vSphere and looking to work with and play around with Proxmox, Proxmox networking may be foreign to work with when trying to compare it with VMware ESXi networking, etc. In this Proxmox networking for vSphere admins post for the community, we will look at the equivalent networking configurations in Proxmox.



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1. Proxmox Linux Bridge equivalent to the VMware vSwitch

The most equivalent network construct out of the box with Proxmox is the default Proxmox Linux bridge in the Proxmox environment. With the Linux Bridge in Proxmox, you establish the initial connectivity to your [Proxmox host](#) with a management IP address. Also, the default Linux Bridge is backed by a physical network adapter(s).

Below:

- Ports/Slaves – This shows the physical network adapter **ens192** assigned to the default Linux bridge
- CIDR – Shows the IP address and mask associated with the Linux bridge



Notes

Shell

System

Network

Certificates

DNS

Hosts

Options

Time

Syslog

Updates

Repositories

Firewall

Disks

LVM

LVM-Thin

Directory

ZFS

Ceph

Replication

Task History

Subscription

ens192	Network Device	Yes	No	No	
vbr0	Linux Bridge	Yes	Yes	No	ens192 10.1.149.155/24

TasksCluster log

Start Time ↓	End Time	Node	User name	Description	Status
Dec 27 02:15:25	Dec 27 02:15:27	pve	root@pam	Update package database	OK
Dec 26 03:36:25	Dec 26 03:36:27	pve	root@pam	Update package database	OK
Dec 25 17:39:00	Dec 25 17:39:00	pve	root@pam	Bulk start VMs and Containers	OK
Dec 25 04:42:16	Dec 25 04:42:20	pve	root@pam	Update package database	OK
Dec 24 05:32:18	Dec 24 05:33:13	pve	root@pam	Update package database	OK

Viewing the default proxmox linux bridge

This is very similar to the default **vSwitch0** created in VMware ESXi right out of an install. As you can see below, we have a physical network adapter backing vSwitch0.



vSwitch0
Type: Standard vSwitch
Port groups: 1
Uplinks: 1

vSwitch Details

MTU	1500
Ports	2060 (2055 available)
Link discovery	Listen / Cisco discovery protocol (CDP)
Attached VMs	0 (0 active)
Beacon interval	1

NIC teaming policy

Notify switches	Yes
Policy	Route based on originating port ID
Reverse policy	Yes
Fallback	Yes

Security policy

Allow promiscuous mode	No
Allow forged transmits	No
Allow MAC changes	No

vSwitch topology

VM Network
VLAN ID: 0
VMkernel ports (1)
vmmk0: 10.3.33.200

Physical adapters
vmmnic0, 1000 Mbps, ...

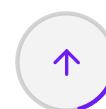
Recent tasks

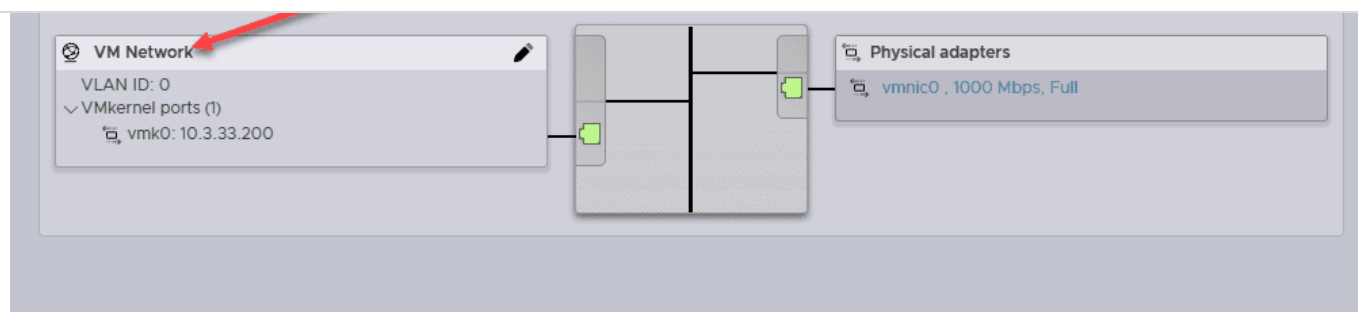
Task	Target	Initiator	Queued	Started	Result	Completed
Refresh Network System	localhost.cloud.local	root	12/27/2023 14:42:56	12/27/2023 14:42:56	Completed successfully	12/27/2023 14:42:56
Update Network Config	localhost.cloud.local	root	12/27/2023 14:42:56	12/27/2023 14:42:56	Completed successfully	12/27/2023 14:42:56
Refresh Network System	localhost.cloud.local	dcui	12/27/2023 14:42:46	12/27/2023 14:42:46	Completed successfully	12/27/2023 14:42:46
Refresh Network System	localhost.cloud.local	root	12/27/2023 14:42:46	12/27/2023 14:42:46	Completed successfully	12/27/2023 14:42:46
Update Network Config	localhost.cloud.local	root	12/27/2023 14:42:41	12/27/2023 14:42:41	Completed successfully	12/27/2023 14:42:43
Refresh Network System	localhost.cloud.local	root	12/27/2023 14:42:18	12/27/2023 14:42:18	Completed successfully	12/27/2023 14:42:18

Vmware esxi default vswitch0

2. Proxmox Linux VLANs equivalent to VMware vSwitch Port Groups

As most vSphere admins are aware, you have the vSwitch0 and the default **VM Network port group** as shown above. Port groups in [vSphere allow you to create VLAN-tagged network labels in vSphere to assign VLAN tags to the virtual machines](#) connected to them.



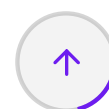


Default vsphere port group on vswitch0

In Proxmox networking with the default Linux Bridge, the port group construct carries over to what is known as a **Linux VLAN**. By default, when you create a VM in Proxmox without any other configuration, you can attach the VM to the default **vmbr0** bridge, which is essentially **VLAN 0** and assumes untagged traffic.

Unlike the [VMware default vSwitch0 and VM Network](#) port group, the default Proxmox Linux Bridge is not VLAN-aware out of the box. You have to enable this.

When you edit the default Linux Bridge, you will see the checkbox **VLAN aware** available on the Linux Bridge properties. Also, you will see [basic networking](#) configurations like the IP address and subnet, gateway for routing, etc. Place a check box in the VLAN aware checkbox.



Making the proxmox linux bridge vlan aware

Now we can apply the configuration. Click the **Apply Configuration** button. Also, in the preview of the Pending changes, you will see the new VLAN bridge-ports configuration set to auto, containing the configuration lines:

```
bridge-ports ens192
bridge-stp off
bridge-vlan-aware yes
bridge-vids 2-4094
```



Apply the vlan aware configuration

This configuration essentially makes the default Linux Bridge able to understand VLANs and VLAN traffic, so we can add Linux VLANs.



Creating a linux vlan

Now we can populate the new Linux VLAN with the appropriate configuration. Once you name the VLAN with the parent **vmbr0** interface, you will see the **VLAN raw device** and **VLAN Tag** greyed out. This essentially says we are creating a new Linux VLAN on the parent Linux Bridge interface, vmbr0.

Under the **Advanced** checkbox, you can set the MTU value in case you are wondering.



Creating the linux vlan from the vmb0 interface

Now that we have created the child VLAN interface on the vmb0 Linux bridge, you can see the **vmb0.333** interface listed now under the network configuration in the navigation tree of **System > network**.

Viewing the newly created linux vlan



Distributed Switch. The [vSphere distributed switch is managed](#) and controlled from vCenter Server. Proxmox doesn't have a centralized management platform like vCenter Server.

Vmware vsphere distributed switch

Proxmox admins would need to manage complex network setups manually with scripting or use third-party tools available for Proxmox for centralized network management.

4. Network Adapters

When it comes to network adapters for a virtual machine or container, both [Proxmox and VMware vSphere support different types of network](#) adapters. These



- E1000
- PCI Passthrough
- VirtIO (Proxmox)

Below you see the vnic options for a virtual machine.

Vmware network adapters

Below are the options when creating a new virtual machine in Proxmox.

Proxmox virtual network adapters



network capabilities. Proxmox does not contain network I/O control features built-in that are found in VMware vSphere. You can use Linux tools in Proxmox to help control network bandwidth.

With Proxmox, you can take advantage of one or many of the following Linux networking tools:

1. Traffic Control (tc):

- tc is a tool in the Linux iproute2 package. It allows you to control the traffic going through your network interfaces. You can create rules for traffic shaping and prioritization. You can also use it for Quality of Service (QoS) rules.

2. iptables:

- iptables can also be used for basic network traffic control. It allows packet filtering and can be combined with tc for more granular control.

3. ethtool:

- ethtool is used for querying and controlling network driver and hardware settings. It can be used to adjust settings like speed, duplex, and autonegotiation on Ethernet interfaces, which can indirectly influence network performance.

4. nftables:

- nftables can be used to set up basic traffic control mechanisms.

5. Wondershaper:

- Wondershaper is a simpler tool [designed to limit the bandwidth of specific network](#) interfaces. It's a good choice for basic bandwidth management.

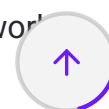
6. VLAN Configuration:

- Configuring VLANs like we discussed above, can help segment network traffic for more efficient network utilization. Linux's native [VLAN configuration tools can be used with Proxmox](#) for this purpose.

7. Network Namespaces:

- Linux network namespaces can be used to isolate network environments for different VMs or containers. This can help manage network traffic and ensure that different services don't interfere with each other's network resources.

8. Monitoring Tools (like iftop, nload, bmon):



information.

9. Cgroups (Control Groups):

- Cgroups, a Linux kernel feature, can be used to limit and prioritize network bandwidth usage among different processes and VMs. While more commonly used for managing CPU and memory resources, cgroups can also be configured to control network I/O.

6. Software-defined networking (SDN)

VMware has a very well-known platform for software-defined networking, known as VMware NSX. The NSX platform is a paid solution on top of vSphere that allows admins to create logical software-defined overlay networks on top of the physical underlay network.

New with Proxmox 8.1 is the introduction of software-defined networking capabilities. You can read the official documentation [here](#). The latest version of Proxmox VE comes with core SDN packages pre-installed. You now have the option for SDN technology in Proxmox VE, allowing admins to create virtual zones and networks (VNets). SDN can also be used for advanced load balancing, NAT, and other features.



Software defined networking in proxmox 8.1

Admins can administer intricate network configurations and multi-tenant environments directly through the web interface at the datacenter level in Proxmox. It allows creating network infrastructure that is more adaptive and responsive and can scale in line with evolving business requirements.

7. Troubleshooting

As you start to work with Proxmox networking, there may be a need for troubleshooting things when networking isn't working correctly. Checking the obvious things like VLANs, VLAN tagging configuration, both in Proxmox, and on your physical network switch are important. If you are using DHCP and DNS to connect to the host, is DHCP handing out the correct IP, and do you have records to resolve the Proxmox host?

Wrapping up



hypervisor. Hopefully, this post will help visitors understand Proxmox networking and the security enhancements available like VLANs, SDN, and others. Proxmox networking isn't so difficult to setup once you understand the equivalents from other virtualization environments you may be familiar with.

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Brandon Lee is the Senior Writer, Engineer and owner at Virtualizationhowto.com, and a 7-time VMware vExpert, with over two decades of experience in Information Technology. Having worked for numerous Fortune 500 companies as well as in various industries, He has extensive experience in various IT segments and is a strong advocate for open source technologies. Brandon holds many industry certifications, loves the outdoors and spending time with family. Also, he goes through the effort of testing and troubleshooting issues, so you don't have to.

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